

SMIT University - Agentic AI Course Overview

Welcome to the SMIT University Agentic AI course notes.

Topic 1: Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that enhances large language model responses by retrieving verified domain knowledge from external knowledge sources, such as vector databases.

Instead of relying entirely on the parametric memory of the model, RAG fetches semantically similar document passages and includes them as context inside the model prompt. This eliminates hallucinations and keeps the AI responses grounded in current, verifiable data.

Vector Databases and Supabase pgvector

Topic 2: Hosted Vector Database Architecture

In production applications, ephemeral or local in-memory vector stores like ChromaDB or FAISS have limitations for enterprise scaling.

Supabase provides a hosted PostgreSQL database with the pgvector extension enabled. PostgreSQL with pgvector provides ACID transactions, relational joins, row-level security, and fast approximate nearest neighbor (ANN) vector indexing via HNSW or IVFFlat.

Document isolation is achieved by tagging each chunk with a document_id metadata field. When similarity search executes, queries filter specifically where metadata->>document_id matches the target document.

Multi-Agent Systems & Model Context Protocol

Topic 3: Model Context Protocol (MCP)

The Model Context Protocol (MCP) developed by Anthropic is an open standard that enables AI models and autonomous agents to interface safely with external tools, repositories, calendars, and mailboxes over JSON-RPC protocols.

In our multi-agent architecture, the LangGraph Supervisor acts as the central router, delegating user goals to specialized worker agents: RAG Agent, GitHub Agent, Google Calendar Agent, and Email Agent.